



Fiona Stanley Fremantle Hospitals Group

Guideline

Administration of Antipsychotic Intramuscular Depot or Long Acting Injections

Reference #: FSH-HW-GUI-0015

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital Fremantle Hospital	Hospital Wide	Medical, Nursing, and Pharmacy

Contents

Scope	1
Contents	1
1. Introduction	2
2. Terminology	2
3. Administration Guideline	2
3.1 General Administration Guidelines	2
3.2 Administration Guidelines for Individual Antipsychotic LAI	3
3.2.1 Risperidone microspheres (Risperdal Consta®)	3
3.2.2 Paliperidone suspension 1 monthly (Invega Sustenna ®)	4
3.2.3 Paliperidone suspension 3 monthly (Invega Trinza ®)	4
3.2.4 Olanzapine (Zyprexa Relprevv®)	5
3.2.5 Aripiprazole (Abilify Maintena®)	6
3.2.6 Flupentixol decanoate (Fluanxol Depot® and Fluanxol Concentrated Depot®), haloperidol decanoate (Haldol Decanoate®) and zuclopenthixol decanoate (Clopixol Depot®)	6
3.3 Missed Doses	7
3.3.1 General principles	7
4. Compliance/Performance Monitoring.....	7
5. Related Policy Documents.....	7

The information provided in this document is based on its relevance to Fiona Stanley Fremantle Hospital Group (FSFHG) only. Due to differences in context, scope of practice and differences in service delivery, FSFHG does not make claim to its relevance or appropriateness for use within other organisations/sites.

7. References	8
8. Authorisation.....	8
Appendix I – Outpatient guide for missed dose administration for clinician administering the LAI	9

1. Introduction

This guideline provides guidance on administration of intramuscular antipsychotic depot or long acting injection, and instructions on how to manage missed doses. The administration of an intramuscular (IM) depot or long acting injection (LAI) is useful for patients who are unable to take oral medication, are non-adherent with oral medication or for those with a preference for an injectable dosage form. When injected deeply into muscle, the medication slowly diffuses into the circulation. Incorrect administration will alter the pharmacokinetics of the LAI which can lead to adverse effects and treatment failure.

Any administration variation outside of this policy will require approval from a consultant psychiatrist and clinical pharmacist consultation.

2. Terminology

Z track technique	Recommended administration technique used to break the needle path which aims to reduce seepage of drug and discomfort at the injection site.
Post Injection Syndrome (PIS)	May develop within an hour of administration of olanzapine LAI. Symptoms are consistent with olanzapine overdose – Oversedation, delirium, extrapyramidal symptoms, dysarthria, ataxia, aggression, dizziness, weakness, hypertension and possible convulsions.

3. Administration Guideline

3.1 General Administration Guidelines

- Verbal consent for administration of LAI needs to be obtained prior to LAI administration. Consider a chaperone when needed.
- Z track technique should always be used to prevent leakage of the medication into subcutaneous tissue.
- Gently aspirate the syringe before delivery of medication to ensure the needle has not entered a blood vessel. If there is blood noticed after aspirating, do not administer and repeat the process with a sterile needle.
- A maximum volume of 3mL can be administered into the gluteal muscle and 2mL into the deltoid muscle. The only exception is the paliperidone 3 monthly

(Paliperidone Trinza®) injection where 2.625mL can be administered into the deltoid muscle.

- For administration that exceeds the recommended volume, the dose is to be split and administered at different injection sites.
- Intramuscular Injection Sites:
- **Ventrogluteal injection site:** This site is the PREFERRED gluteal injection site and it is located midway between the two bone structures of the hip and head of femur as:
 - There are bony landmarks that are considered easy to palpate
 - Greater thickness of muscle than the dorsogluteal site
 - It has a thin layer of subcutaneous fat which facilitate the LAI being effectively administered into the muscle
 - The site is relatively free of nerves and blood vessels reducing the potential for injury and pain.
- **Dorsogluteal injection site:** This site is NOT the preferred gluteal injection site and is located in the upper outer quadrant of the buttock, in the superior lateral aspect of the gluteal muscles as:
 - The presence of major nerves such as the sciatic nerve and blood vessels in the area
 - Potential thick layer of adipose tissue increasing the risk of the LAI being administered into adipose tissue rather than muscle.
- **Deltoid injection site:** This site is located at or immediately below the midpoint of the deltoid muscle and above the level of the axilla.
 - Pain is reported as the most frequently reported complications associated with deltoid injection. The deltoid muscle has the greatest blood flow of the muscles thus provides a more rapid onset of LAI but a shorter duration of action for the LAI.¹

3.2 Administration Guidelines for Individual Antipsychotic LAI

3.2.1 Risperidone microspheres (Risperdal Consta®)

- Administer by deep IM injection into the DELTOID muscle or GLUTEAL muscle using the appropriate safety needle provided by the manufacturer.
 - Alternate injections between left and right sides.
- Remove the dose from the refrigerator approximately 30 minutes before reconstitution and allow it to reach room temperature.
 - It is recommended that the dose is administered immediately after reconstitution with the supplied diluent.
- Use the specified gauge of safety needle that is provided by the manufacturer based on the administration site:
 - For deltoid administration:

- Use the 21 gauge 1 inch needle
 - For gluteal administration:
 - Use the 20 gauge 2 inch needle
- For high dose risperidone (75mg):
 - Using two boxes of the 37.5mg Risperdal Consta® injection, use one of the 2mL diluents supplied to reconstitute one of the vials. Discard the second 2mL diluent.
 - Withdraw this solution and use to reconstitute the second 37.5mg vial.
 - Withdraw the total solution into the syringe and administer using the appropriate needle for the injection site.
 - Give as a single deep IM injection.

3.2.2 Paliperidone suspension 1 monthly (Invega Sustenna ®)

- **Initiation (Day 1) and Loading (Day 8):**
 - Administer by deep IM DELTOID injection alternating between the left and right sides.
 - Use the specified gauge of safety needle based on the patient's body weight that is provided by the manufacturer for this deltoid administration:
 - If < 90kg use 23 gauge 1 inch needle
 - If ≥ 90kg use 22 gauge 1.5 inch needle
- **Maintenance:**
 - Following the second dose, monthly doses can be administered in either the deltoid or gluteal muscle alternating between the left and right sides.
 - The recommended needle size for administration of paliperidone into the deltoid muscle is determined by the patient's body weight:
 - If < 90kg use 23 gauge 1 inch needle
 - If ≥ 90kg use 22 gauge 1.5 inch needle
- The recommended needle size for administration of paliperidone into upper outer quadrant gluteal muscle:
 - Use 22 gauge 1.5 inch needle

3.2.3 Paliperidone suspension 3 monthly (Invega Trinza ®)

- Prior to administration, shake the syringe vigorously for at least 15 seconds with the syringe tip pointing up to ensure a homogenous suspension.
- Must be administered by single slow deep IM DELTOID or GLUTEAL injection alternating between the left and right sides.
 - **Deltoid Administration:**

- Use the needle provided by the manufacturer for administration of paliperidone into the centre of the deltoid.
- The appropriate size will be based on the patient's body weight:
- If < 90kg use 22 gauge 1 inch thin wall needle
- If ≥ 90kg use 22 gauge 1.5 inch thin wall needle
- **Gluteal Administration:**
 - Use the needle provided by the manufacturer for administration of paliperidone into the upper-outer quadrant of the gluteal muscle.
 - The needle size recommended is regardless of body weight:
 - Use 22 gauge 1.5 inch thin wall needle
 - Administer within 5 minutes or repeat the 15 seconds of shaking.
- Incompletely administered dose:
 - Contact pharmacist for clinical advice.

Do not re-inject the dose remaining in the syringe and do not administer another dose.

3.2.4 Olanzapine (Zyprexa Relprevv®)

- Must be administered by deep IM GLUTEAL injection ONLY, alternating between the left and right sides. Deltoid injections are associated with higher risk of post-injection syndrome (PIS)
- Once reconstituted, the product should be used as soon as possible. The reconstituted product may be stored for up to 6 hours at room temperature (25°C) before being used.
- Use the appropriate specified gauge safety needle that is provided by the manufacturer:
 - In non-obese patients use 19 gauge 1.5 inch needle
 - In obese patients use 19 gauge 2 inch needle
- **Monitoring:**
 - PIS occurred at a rate of 0.07% in premarketing studies.
 - Symptoms of PIS are consistent with olanzapine overdose and in most cases will occur within 1 hour of administration.
 - These include sedation and delirium (disorientation, confusion, agitation, anxiety and other cognitive impairment), extrapyramidal symptoms, dysarthria (slurred speech), ataxia (staggering, uneven gait), aggression, dizziness, weakness, hypertension and convulsion.
- The following monitoring is required:

- Baseline set of physiological observations must be taken prior to administration and documented on the Adult Observation and Response chart (AORC)
- The patient is to be assessed every 30 minutes for 2 hours post injection for symptoms of PIS, orientation and level of alertness. This is to be documented on the AORC and in the medical record.
- The period of monitoring may be extended to 3 hours or longer post injection dependent upon medical advice
- Prior to discharge a complete set of vital observations must be documented on the AORC and in the medical record
- Patient is informed not to drive home or operate heavy machinery for 24 hours post administration of olanzapine.

3.2.5 Aripiprazole (Abilify Maintena®)

- Must be administered by deep IM DELTOID or GLUTEAL injection alternating between the left and right sides.
- The reconstituted suspension should be used immediately, but may be stored below 25°C for up to 4 hours. Shake the vial vigorously for at least 60 seconds to re-suspend prior to injection.
- Use the appropriate specified gauge safety needle that is provided by the manufacturer:
 - **Deltoid administration:**
 - In non-obese patients use 23 gauge 1 inch needle
 - In obese patients use 22 gauge 1.5 inch needle
 - **Gluteal administration:**
 - In non-obese patients use 22 gauge 1.5 inch needle
 - In obese patients use 21 gauge 2 inch needle

3.2.6 Flupentixol decanoate (Fluanxol Depot® and Fluanxol Concentrated Depot®), haloperidol decanoate (Haldol Decanoate®) and zuclopenthixol decanoate (Clopixol Depot®)

- Flupentixol and Haloperidol LAI are only registered to be administered in the GLUTEAL muscle. The site for zuclopenthixol decanoate LAI is not specified in the product information but GLUTEAL administration is recommended.
- Administer by deep IM injection alternating between the left and right muscle.

3.3 Missed Doses

3.3.1 General principles

- The maximum accepted delay and the need for dosing interventions relates to the dose and frequency of the LAI prescribed. **All** early administration of LAI will require approval from the treating medical officer.
- For all inpatient missed dose of antipsychotic LAI, contact the treating medical team and clinical pharmacist for advice.
- All outpatient missed dose of antipsychotic LAI will be managed as per the following protocol:
 - Ascertain the date of last injection
 - Calculate the amount of time that has passed since the last injection. Refer to the individual recommendations below to determine if the LAI can be administered.
 - If it is outside the timeframe outlined in this policy:
 - Contact the treating medical officer

3.3.1.3 Refer to [Appendix I](#) for guidance on missed dose for individual medications.

4. Compliance/Performance Monitoring

- Compliance against this policy will be evaluated with routine incident review processes.
- Clinical educators in the Mental Health Unit will be responsible for ensuring all nursing staff are familiar with and adhere to the direction given in this guideline.

5. Related Policy Documents

Department of Health

- [The WA Medicines Formulary](#)

FSFHG

- [Medication Administration](#) (FSFH-HW-POL-0022)

6. Related Standards

NSQHS v2

- Medication Safety
- Partnering with consumers
- Communicating for Safety

7. Bibliography

1. Cocoman A, Murray J. Intramuscular injections: a review of best practice for mental health nurses. *Journal of Psychiatric and Mental Health Nursing*, 2008;15;424-434
2. Arslan GG, Özden D. Creating a change in the use of ventrogluteal site for intramuscular injection. *Patient Prefer Adherence*. 2018;12:1749-1756. Published 2018 Sep 13. doi:10.2147/PPA.S168885
3. Bazire, S. (2020/21). *Psychotropic Drug Directory*. London, UK: Park Communications Limited
4. MIMs Online. (2021). Product Information. Retrieved from:
<https://www-mimsonline-com-au.smhslibresources.health.wa.gov.au/Search/Search.aspx>
5. Buckley N, editor. (2021). *Australian Medicines Handbook*. Adelaide: Australian Medicines Handbook; 2021 [online] Accessed 08/2021
6. Therapeutic Guidelines Pty Ltd. (2018). *Psychotropic Guidelines*. Retrieved from: <http://online.tg.org.au.smhslibresources.health.wa.gov.au/ip/desktop/index.htm>
7. SHPA (2021) *The Australian Injectable Drugs Handbook*. 8th ed. [online] Accessed 08/2021. Available from: https://aidh-hcn-com-au.smhslibresources.health.wa.gov.au/browse/about_aidh
8. Taylor D, Barnes T, Young A. *The Maudsley Prescribing Guidelines in Psychiatry* [online] 2018. Accessed 09/2018. Available from: <https://ebookcentral-proquestcom.smhslibresources.health.wa.gov.au/lib/semhs/detail.action?docID=5349044>
9. Hunter J. Intramuscular injection techniques. *Nursing Standard* (through 2013) 2008 Feb;22(24):35-40.

8. Authorisation

EXECUTIVE SPONSOR: Nurse Director, Service 2					
ENDORSEMENT PROCESS: Hospital wide					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	03/2015	FSH Senior Clinical Pharmacist	DTC	HEC	03/2018
2	11/2018	FSH Senior Clinical Pharmacist	DTC, NMPGC	HEC Policy and Procedure Committee	11/2021
3	05/2022	FH Mental Health Lead Pharmacist	DTC, FSFGH-MH SQR, FH Pharmacy, FSH Pharmacy, FH-MH Medical, FSH-MH Medical, FH-MH Nursing, FSH-MH Nursing	FSFH Policy Committee	08/2026

Appendix I – Outpatient guide for missed dose administration for clinician administering the LAI

Antipsychotic LAI	Administration before due date	Administer after the due date without first contacting treating medical officer*	Next Due Date
Aripiprazole	Contact Medical Officer	3 days	Resume the same due date from the missed dose
Flupentixol	Contact Medical Officer	2 days	Calculate from date administered
Haloperidol	Contact Medical Officer	2 days	Calculate from date administered
Olanzapine	Contact Medical Officer	2 days	Calculate from date administered
Paliperidone 1 monthly (Invega Sustenna®)	Contact Medical Officer	3 days	Calculate from date administered
Paliperidone 3 monthly (Invega Trinza®)	Contact Medical Officer	7 days	Resume the same due date from the missed dose
Risperidone	Contact Medical Officer	2 days	Calculate from date administered
Zuclopenthixol decanoate	Contact Medical Officer	2 days	Calculate from date administered

* The treating medical officer needs to be notified and the late administration will be recorded in the medical records